

HERO / F-HERO — CLI cheat sheet (Aug 2026)

Regenerate raw `--help` dumps:

```
# From repo root - writes population_sandbox/cli_help_dump.txt
./scripts/dump_hero_cli_help.sh
```

Paste into Word: open `cli_help_dump.txt`, or copy sections from this file.

Three orthogonal knobs (do not conflate)

Knob	Flag	Default	Meaning
Y labeling	<code>--y-draft-mode</code>	<code>ever</code>	<code>ever</code> : every PS row of a draftee gets Y=1. <code>season</code> : Y=1 only on last college PS; earlier PS kept with Y=0.
Panel rows	<code>--panel-rows</code>	<code>all-ps</code>	<code>all-ps</code> : all player-seasons (fr/so/jr/sr). <code>last-ps</code> : one row per athlete (max season).
Roster x-axis (HERO only)	<code>--roster-x</code>	<code>poolq_loo</code>	<code>poolq_loo</code> : LOO teammate mean (self out). <code>poolq</code> : team-season mean (self in).

2x2: Y × panel (HERO + F-HERO)

	<code>panel-rows all-ps</code>	<code>panel-rows last-ps</code>
<code>y-draft-mode ever</code>	Canonical deck HERO	Final-season cross-section
<code>y-draft-mode season</code>	Full panel; Y=1 last PS only	Final-season + season-Y

Retired (hidden): `--last-season-only` → use `--panel-rows last-ps`. `--all-seasons` → use `--panel-rows all-ps`.

Script 1 — `pass_a_empirical_bundle.py` (HERO)

What it does: Left = ability ventiles; right = roster-pressure hero; standalone HERO PNG + CSVs + provenance JSON.

Usage:

```
python sports/scripts/pass_a_empirical_bundle.py [options]
```

Flag	Choices / default	Notes
<code>--season-min</code> / <code>--season-max</code>	2011 / 2021	Season window
<code>--n-bins</code>	16	Hero + ability bin count
<code>--poolq-binning</code>	<code>quantile</code> · <code>equal_width</code>	Right panel only
<code>--roster-x</code>	<code>poolq_loo</code> · <code>poolq</code>	LOO vs team mean
<code>--min-minutes</code>	20	Playing-time floor
<code>--min-team-season-games</code>	10 (<code>mg10</code>)	Box QC; use <code>0</code> for mg0 audit only
<code>--winsor-lo</code> / <code>--winsor-hi</code>	0.01 / 0.99	On roster-x column
<code>--y-draft-mode</code>	<code>ever</code> · <code>season</code>	Y labeling
<code>--panel-rows</code>	<code>all-ps</code> · <code>last-ps</code>	Which rows
<code>--dft</code>	off	+DFT team filter
<code>--output-root</code>	<code>pass_a/</code> or sandbox	Sandbox: <code>.../population_sandbox/hero</code>
<code>--output-tag</code>	optional	Extra filename slug
<code>--perf-metric</code>	<code>ppm</code> (+ obpm/bpm/...)	Ability + pool source

Canonical (deck):

```
python sports/scripts/pass_a_empirical_bundle.py \
--output-root 3-Master_Plan/re_entry/HEROs_and_PASSes/population_sandbox/hero
```

Season-Y + full panel + LOO:

```
python sports/scripts/pass_a_empirical_bundle.py \
--y-draft-mode season --panel-rows all-ps \
--output-root 3-Master_Plan/re_entry/HEROs_and_PASSes/population_sandbox/hero
```

Team poolq (not LOO):

```
python sports/scripts/pass_a_empirical_bundle.py \
--roster-x poolq \
--output-root 3-Master_Plan/re_entry/HEROs_and_PASSes/population_sandbox/hero
```

Filename tokens: `HERO_[poolq_]q16_allt_min20_mg10_11_21[_season_y][_last_ps].png`

Script 2 — `pass_a_congestion_conditional.py` (F-HERO / CCT)

What it does: Conditional draft-rate plots. Alex Slide 10 = `--plot fixed_ai_tj_knbins`.

Usage:

```
python sports/scripts/pass_a_congestion_conditional.py --plot fixed_ai_tj_knbins [options]
```

Flag	Choices / default	Notes
<code>--plot</code>	<code>fixed_ai_tj_knbins</code> (F-HERO)	Also: <code>matched_pond</code> , <code>p1_grid</code> , ...
<code>--season-min</code> / <code>--season-max</code>	2011 / 2021	
<code>--min-minutes</code>	20	
<code>--min-team-season-games</code>	10	
<code>--ai-top-pct</code>	7 (typical)	Top X% Å band; vs <code>--ai-lo</code> / <code>--ai-hi</code>
<code>--dft</code>	off	Slide 10 uses +DFT
<code>--y-draft-mode</code>	<code>ever</code> · <code>season</code>	Same semantics as HERO
<code>--panel-rows</code>	<code>all-ps</code> · <code>last-ps</code>	Same semantics as HERO
<code>--tj-binning</code>	<code>piecewise_tail</code>	F-HERO <code>pw4p7</code>
<code>--tj-n-low</code> / <code>--tj-n-high</code>	4 / 7	Piecewise T _j bins
<code>--p2b-single</code>	off	One spec only (no bundle)
<code>--out-dir</code>	<code>basic_data_plots/</code>	Sandbox: <code>.../population_sandbox/fhero</code>

Canonical F-HERO:

```
python sports/scripts/pass_a_congestion_conditional.py \
--plot fixed_ai_tj_knbins --p2b-single --dft \
--min-minutes 20 --ai-top-pct 7 --tj-n-low 4 --tj-n-high 7 \
--out-dir 3-Master_Plan/re_entry/HEROs_and_PASSes/population_sandbox/fhero
```

13–21 sweep: add `--season-min 2013 --season-max 2021`.

Script 3 — `pass_a_hero_sensitivity_plots.py`

What it does: Gallery of labeled sensitivity PNGs under `pass_a/sensitivity/` (includes LOO vs `pool_mean` grid).

```
python sports/scripts/pass_a_hero_sensitivity_plots.py --quick
python sports/scripts/pass_a_hero_sensitivity_plots.py --perf-metric obpm
```

Flag	Notes
<code>--quick</code>	Smoke subset (~8 specs)
<code>--perf-metric</code>	Default <code>ppm</code>

Script 4 — `compare_hero_mg_sandbox.py`

What it does: Ventile diff table mg0 vs mg10 in sandbox (runs mg0 if missing).

```
python sports/scripts/compare_hero_mg_sandbox.py
```

No CLI flags — paths fixed to `population_sandbox/hero/`.

Output roots (sandbox)

Folder	Script	--output-root / --out-dir
population_sandbox/hero/	pass_a_empirical_bundle.py	--output-root ../hero
population_sandbox/fhero/	pass_a_congestion_conditional.py	--out-dir ../fhero

Every HERO PNG has `*_provenance.json` with full footer spec.

COMPASS · update when CLI changes · run `./scripts/dump_hero_cli_help.sh` to refresh raw help